

G37 AI & ML – Individual Questions on EDA, Statistics, Linear Regression, KNN & Logistic Regression

1. Explain how you would detect and handle outliers during Exploratory Data Analysis when the dataset has skewed distributions.
2. During EDA, how would you decide whether a feature should be transformed using log or square root transformation?
3. What visualizations would you choose to understand relationships between multiple numerical variables, and why?
4. How can missing data patterns discovered during EDA influence model selection?
5. Explain the role of correlation matrices in EDA and one limitation of relying on them.
6. How would you perform EDA differently for time-series data compared to tabular data?
7. Describe how EDA can help identify data leakage before model training.
8. Explain how boxplots and violin plots convey different information in EDA.
9. What statistical measures would you use to summarize a dataset with extreme outliers, and why?
10. Explain the difference between population variance and sample variance with a real-world example.
11. How does the Central Limit Theorem justify the use of parametric tests?
12. When would median be preferred over mean as a measure of central tendency?
13. Explain covariance and how it differs from correlation.
14. What is the significance of degrees of freedom in statistical inference?
15. How does multicollinearity affect statistical interpretation?
16. Explain Type I and Type II errors using a machine learning context.
17. What assumptions must hold true for linear regression to be valid?
18. How do you interpret the coefficients of a multiple linear regression model?
19. Explain the concept of residuals and their importance in linear regression diagnostics.
20. What is heteroscedasticity and how can it be detected?
21. How does regularization improve linear regression models?
22. Explain the difference between R-squared and Adjusted R-squared.
23. When would polynomial regression be preferred over simple linear regression?
24. How does feature scaling impact linear regression performance?
25. Explain the intuition behind distance metrics used in KNN.

26. How does the choice of K affect bias and variance in KNN?
27. Why is KNN considered a lazy learning algorithm?
28. How does KNN handle multi-class classification problems?
29. Explain the impact of feature scaling on KNN performance.
30. What challenges does KNN face with large datasets?
31. How would you choose an optimal value of K?
32. Explain how KNN can be adapted for regression problems.
33. What is the curse of dimensionality and how does it affect KNN?
34. Explain the sigmoid function and its role in logistic regression.
35. How does logistic regression differ from linear regression in output interpretation?
36. What is the log-odds concept in logistic regression?
37. Explain maximum likelihood estimation in the context of logistic regression.
38. How do you interpret coefficients in logistic regression?
39. What is the decision boundary in logistic regression?
40. How does class imbalance affect logistic regression performance?
41. Explain the role of regularization in logistic regression.
42. What evaluation metrics are more suitable for logistic regression than accuracy?
43. How does logistic regression handle non-linear relationships?
44. Explain the difference between binary and multinomial logistic regression.
45. How would you diagnose overfitting in logistic regression?
46. Design an EDA strategy for a healthcare dataset containing patient vitals.
47. How would you statistically compare two datasets to check if they come from the same distribution?
48. Explain how interaction terms work in linear regression.
49. What happens if irrelevant features are included in a linear regression model?
50. How does weighted KNN differ from standard KNN?
51. Explain why KNN is sensitive to noisy data.
52. How can dimensionality reduction improve KNN performance?
53. What assumptions does logistic regression make about the data?
54. Explain ROC curve and AUC in the context of logistic regression.
55. How would you perform EDA on a dataset with both images and tabular features?
56. Explain sampling bias and its statistical consequences.

57. How does multicollinearity impact linear regression coefficients?
58. Explain why linear regression may fail for classification tasks.
59. How does distance weighting change prediction behavior in KNN?
60. What is threshold tuning in logistic regression and why is it important?
61. How would you justify model choice between KNN and logistic regression?
62. Explain the difference between generative and discriminative models using logistic regression.
63. How can EDA help improve feature engineering for ML models?
64. Explain one real-world use case where logistic regression is preferred over complex models.